LECTURE 16

Theorem 0.1 *Let $\{A_n\}$ and $\{B_n\}$ be countable partitions of Ω from $\mathcal{F}$ and $\{a_n\}$, $\{b_n\}$ be a sequence of real numbers such that a_n's are distinct and b_n's are not necessarily distinct. Then if*

$$\sum_{n=1}^{\infty} a_n I_{A_n} \;=\; \sum_{n=1}^{\infty} b_n I_{B_n}$$

then

$$\sum_{n=1}^{\infty} a_n P(A_n) \;=\; \sum_{n=1}^{\infty} b_n P(B_n)$$

provided l.h.s. series converges absolutely.

Proof. Using the hypothesis given it follows that

1. Each B_m intersect atmost one A_n. To see this, firstly if B_m intersect say A_n and A_k, then

   $$B_m \cap A_n \neq \emptyset \;\Rightarrow\; a_n = b_m,\; B_m \cap A_k \neq \emptyset \;\Rightarrow\; a_k = b_m.$$

   i.e., $a_n = a_k$ and hence $A_n = A_k$.

2. For each B_m, there exists A_n such that $B_m \subseteq A_n$. This follows from:

   $$\begin{aligned} B_m \qquad &= \qquad \bigcup_{n=1}^{\infty} (B_m \cap A_n) \\ \text{using (1)} \Rightarrow \quad &\text{for some } n, \quad B_m = B_m \cap A_n. \end{aligned}$$

Thus we conclude that for each $n \geq 1$, either $B_m \subseteq A_n$ or $B_m \cap A_n = \emptyset$ for all $m \geq 1$. Hence, for each $n \geq 1$, set

$$I_n \;=\; \{m \geq 1 | A_n \cap B_m \neq \emptyset\}\,.$$

Then clearly

$$A_n \;=\; \cup_{m \in I_n} B_m,\; n \geq 1\,.$$

Also if $m \in I_n$ then $a_n = b_m$. Therefore (using absolute summability of the series)

$$\begin{aligned}\sum_{m=1}^{\infty} b_m P(B_m) &= \sum_{n=1}^{\infty} \sum_{m \in I_n} b_m P(B_m) \\ &= \sum_{n=1}^{\infty} \sum_{m \in I_n} a_n P(B_m) \\ &= \sum_{n=1}^{\infty} a_n P(A_n)\,.\end{aligned}$$

This completes the proof.

Definition 0.1 *Let X be a discrete random variable represented by $\{(A_n,\ a_n) \,|\, n \geq 1\}$, where $\{A_n\} \subseteq \mathcal{F}$ is a partition of Ω. Then expectation of X denoted by EX is defined as*

$$EX \;=\; \sum_{n=1}^{\infty} a_n P(A_n)\,,$$

provided the right hand side series converges absolutely.

Example 0.1 *Let X be a Bernoulli(p) random variable given by*

$$X \;=\; I_A\,,\ \text{where } A \;=\; \{X \;=\; 1\}\,.$$

$$EX \;=\; P(A) \;=\; p\,.$$

Example 0.2 *Let X be a Binomial$(n,\, p)$ random variable given by*

$$X \;=\; \sum_{k=0}^{n} k\, I_{A_k},$$

where $\{A_k\} \subseteq \mathcal{F}$ is a partition of the sample space Ω.

$$\begin{aligned}EX &= \sum_{k=0}^{n} kP(A_k) \;=\; \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=1}^{n} n \binom{n-1}{k-1} p^k (1-p)^{n-k} \;=\; np\,.\end{aligned}$$

Here we used the identity

$$k \binom{n}{k} \;=\; n \binom{n-1}{k-1}\,.$$

Example 0.3 *Let X be a Poisson (λ) random variable given by*

$$X \;=\; \sum_{n=0}^{\infty} n\, I_{A_n},$$

where $\{A_n\} \subseteq \mathcal{F}$ is a partition of the sample space Ω.

$$EX \;=\; \sum_{n=0}^{\infty} n\, e^{-\lambda}\, \frac{\lambda^n}{n!} \;=\; \lambda\,.$$

Example 0.4 *Let X be a Geometric (p) random variable given by*

$$X \;=\; \sum_{n=1}^{\infty} n\, I_{A_n},$$

where $\{A_n\} \subseteq \mathcal{F}$ is a partition of the sample space Ω. [1]

$$EX \;=\; \sum_{n=1}^{\infty} n\, p\, (1-p)^{n-1} \;=\; \frac{p}{(1-(1-p))^2} \;=\; \frac{1}{p}\,.$$

Theorem 0.2 *(Properties of expectation) Let X and Y be discrete random variables with finite means. Then*

(i) If $X \geq 0$, then $EX \geq 0$.

(ii) For $a \in \mathbb{R}$

$$E(aX + Y) \;=\; aEX + EY\,.$$

Proof. (i) Let $\{(A_n,\ a_n) | n \geq 1\}$ be a representation of X. Then $X \geq 0$ implies $a_n \geq 0$ for all $n \geq 1$. Hence

$$EX \;=\; \sum_{n=1}^{\infty} a_n P(A_n) \;\geq\; 0\,.$$

[1] Another 'model' for geometric random variable is the one which represents the number of failures before the first success, then

$$X \;=\; \sum_{n=0}^{\infty} n\, I_{A_n},\ P(A_n) = (1-p)^n p, n \geq 0.$$

Compute EX in this case.

(ii) Let Y has a representation $\{(B_n, b_n) \,|\, n \geq 1\}$. Now set

$$C_{nm} \;=\; A_n \cap B_m,\; n, m \geq 1, a_{nm} \;=\; a_n, m \geq 1, b_{nm} \;=\; b_m, n \geq 1$$

Then $\{C_{nm} : n, m \geq 1\}$ is a partion of Ω and

$$X \;=\; \sum_{n,m=1}^{\infty} a_{nm} I_{C_{nm}},\; Y \;=\; \sum_{n,m=1}^{\infty} b_{nm} I_{C_{nm}},$$

i.e., X, Y are represented using a common partition of Ω. Therefore

$$aX + Y \;=\; \sum_{n,m=1}^{\infty} (aa_{nm} \,+\, b_{nm})\, I_{C_{nm}}\,.$$

Hence

$$\begin{aligned}
a\,EX \;+\; EY \;&=\; a \sum_{n=1}^{\infty} a_n\, P(A_n) \;+\; \sum_{m=1}^{\infty} b_m\, P(B_m) \\
&=\; a \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_n\, P(A_n \cap B_m) \;+\; \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} b_m\, P(A_n \cap B_m) \\
&=\; a \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{nm}\, P(C_{nm}) \;+\; \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} b_{nm}\, P(C_{nm}) \\
&=\; \sum_{n,m=1}^{\infty} (aa_{nm} \,+\, b_{nm})\, P(C_{nm}) \\
&=\; E(aX + Y)\,.
\end{aligned}$$

Definition 0.2 *(Simple random variable) A random variable is said to be simple if it is of the form*

$$\sum_{n=1}^{N} a_n I_{A_n},$$

for some $\infty > N \geq 1, a_n \in \mathbb{R}$, and partition $\{A_n\} \subseteq \mathcal{F}$ of Ω.

Theorem 0.3 *Let X be random variable in $(\Omega, \mathcal{F}, P)$ such that $X \geq 0$, then there exists a sequence of simple random variables $\{X_n\}$satisfying*

(i) For each $n \geq 1$, $X_n \geq 0,\; X_n \leq X_{n+1} \leq X$.

(ii) For each $\omega \in \Omega,\; X_n(\omega) \to X(\omega)$ as $n \to \infty$.

Proof. For $n \geq 1$, define simple random variable X_n as follows:

$$X_n(\omega) \;=\; \begin{cases} \dfrac{k}{2^n} & \text{if} \quad \dfrac{k}{2^n} \leq X(\omega) < \dfrac{k+1}{2^n} \;, k = 0, \cdots, n2^n - 1 \\ 0 & \text{if} \quad X(\omega) \geq n \,. \end{cases}$$

Then X_n's satisfies the following:

- if $X_n(\omega) = \frac{k}{2^n}$ (i.e. $X(\omega) \in [\frac{k}{2^n},\ \frac{k+1}{2^n})$), then $X(\omega) \in [\frac{2k}{2^{2n}},\ \frac{2k+1}{2^{2n}}) \cup [\frac{2k+1}{2^{2n}},\ \frac{2k+2}{2^{2n}})$. Hence $X_{n+1}(\omega) = \frac{k}{2^n}$ or $= \frac{2k+1}{2^{2n}}$.
- For $X(\omega) < n$,
$$|X_n(\omega) - X(\omega)| \leq \frac{1}{2^n}.$$

The above implies that

$$X_n \;\leq X_{n+1},\ n \geq 1$$
$$\lim_{n\to\infty} X_n(\omega) \;=\; X(\omega),\ \omega \in \Omega \,.$$

Lemma 0.1 *Let X be a non negative random variable and $\{X_n\}$ be a sequence of simple random variables satisfying (i) and (ii) of Theorem 0.3. Then $\lim_{n\to\infty} EX_n$ exists and is given by*

$$\begin{aligned} \lim_{n\to\infty} EX_n \;&=\; \sup\{EY \,|\, Y \text{ is simple and } Y \leq X\} \\ &=\; \sup\{EY \,|\, Y \text{ is simple and } Y \leq X \text{ a.s. }\} \end{aligned}$$

Proof. (Reading exercise) Since $X_n \leq X_{n+1}$, we have $EX_n \leq EX_{n+1}, n \geq 1$ (see exercise). Hence $\lim_{n\to\infty} EX_n$ exists. Also since X_n's are simple, clearly,

$$EX_n \;\leq\; \sup\{EY \,|\, Y \text{ is simple and } Y \leq X\},\ n \geq 1.$$

Therefore

$$\lim_{n\to\infty} EX_n \;\leq\; \sup\{EY \,|\, Y \text{ is simple and } Y \leq X\} \,.$$

Hence to complete the proof, it suffices to show that for Y simple and $Y \leq X$,

$$EY \;\leq\; \lim_{n\to\infty} EX_n \,.$$

Let

$$Y \;=\; \sum_{k=1}^{m} a_k \, I_{A_k} \,,$$

where $\{A_k \,|\, k = 1, \ldots, m\}$ is a partition of Ω. Fix $\epsilon > 0$, set for $k \geq 1$ and $n \geq 1$,

$$A_{kn} \;=\; \{\omega \in A_k \,|\, X_n(\omega) \;\geq\; a_k \,-\, \epsilon\}\,.$$

Then

$$X_n \geq \sum_{k=1}^{m} (a_k - \varepsilon)\, I_{A_{kn}}.$$

Also, since $X_n \nearrow X$, it follows that [2]

$$A_{kn} \nearrow A_k,\; k = 1, 2, \cdots, m.$$

Hence

$$EX_n \;\geq\; \sum_{k=1}^{m} (a_k - \epsilon) P(A_{kn})\,. \tag{0.1}$$

Using continuity property of probability, we have

$$\lim_{n\to\infty} P(A_{kn}) \;=\; P(A_k),\; 1 \leq k \leq m\,.$$

Now let, $n \to \infty$ in (0.1), we get

$$\lim_{n\to\infty} EX_n \geq \;\sum_{k=1}^{m} (a_k - \epsilon) P(A_k) \;=\; EY - \epsilon\,.$$

Since, $\epsilon > 0$ is arbitrary, we get

$$\lim_{n\to\infty} EX_n \geq \;EY\,.$$

This completes the proof.

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[2] Since $X_n \leq X_{n+1},\; n \geq 1$, we have for each $k \geq 1,\; A_{k\,n} \subseteq A_{k\,n+1}, n \geq 1\,.$

$$\begin{array}{rcl} \omega \in A_k & \Longrightarrow & X(\omega) \geq a_k \;\Longrightarrow\; \lim_{n\to\infty} X_n(\omega) \geq a_k \\ & \Longrightarrow & X_{n_0}(\omega) \geq a_k - \epsilon \text{ for some } n_0 \\ & \Longrightarrow & \omega \in A_{kn_0} \subseteq \cup_{n=1}^{\infty} A_{kn}\,. \end{array}$$

Therefore $\cup_{n=1}^{\infty} A_{kn} \;=\; A_k,\; 1 \leq k \leq m\,.$